

PHYS 217 Electricity and Magnetism I

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1 Vector Analysis

1.1 Vector Algebra

Definition 1.1. A **vector** is a quantity that has both direction and magnitude.

Definition 1.2. A **scalar** is a quantity that only has magnitude.

1.1.1 Vector Operations

There are four vector operations:

Addition:

- $\mathbf{A} + \mathbf{B}$ is the vector from the start of $\mathbf{A}$ to the end of $\mathbf{B}$
- Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- Subtracting a vector is just adding its opposite: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Multiplication by Scalar:

- $a\mathbf{A}$ is a vector with the same direction as $\mathbf{A}$ but a times the magnitude
- Distributive: $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$

Dot Product (Inner Product or Scalar Product):

- $\mathbf{A} \cdot \mathbf{B} \equiv AB \cos \theta$
- Results in scalar
- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A and the projection of $\mathbf{A}$ along $\mathbf{B}$

Cross Product of Two Vectors:

- $\mathbf{A} \times \mathbf{B} \equiv AB \sin \theta \hat{\mathbf{n}}$ where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. The direction
- is determined by the **right hand rule**, where you curl your right hand from $\mathbf{A}$ to $\mathbf{B}$ and take the direction of your thumb
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$
- Not commutative: $\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$
- Result is a **pseudovector** and behaves like a vector under rigid transformations like rotations, but not under certain nonrigid transformations. Notably, $-\mathbf{A} \times -\mathbf{B} = \mathbf{A} \times \mathbf{B}$
- Geometrically, $|\mathbf{A} \times \mathbf{B}|$ is the area of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$

When working in Cartesian coordinates, vectors can be expressed in terms of **basis vectors**:

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

with A_x , A_y , and A_z being the **components** of $\mathbf{A}$.

Vector operations can be done with the components, but the reader should already know how to do this.

There are two types of triple vector products:

Scalar Triple Product:

- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$
- Volume of parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$

Vector Triple Product:

- **BAC-CAB rule:** $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

1.1.2 Nomenclature

Definition 1.3. The **position vector** of a point is the vector to that point from the origin:

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

Definition 1.4. The **infinitesimal displacement vector** from (x, y, z) to $(x + dx, y + dy, z + dz)$ is:

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$$

Definition 1.5. A **separation vector** is the vector from a **source point** to a **field point**:

$$\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'$$

1.1.3 Vector Transformations

Coordination transformation from rotation about an arbitrary axis can be represented by the use of a **rotation matrix**:

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

or more concisely:

$$\bar{A}_i = \sum_{j=1}^3 R_{ij} A_j$$

Definition 1.6. A transformation is **unitary** if it does not change the magnitude of the object on which it operates.

If rotation matrix $\overleftrightarrow{R}$ is unitary, then:

$$\sum_i R_{ij} R_{ik} = \delta_{jk}$$

where:

$$\delta_{jk} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the Kronecker delta.

Definition 1.7. **Parity** is a property related to how a system is affected by coordinate inversions. If a system stays the same after a parity transformation, it is said to have even parity. If it is inverted, it is said to have odd parity.

1.1.4 Dyadics and Tensors

Definition 1.8. A **tensor** is something that transforms like a tensor.

Definition 1.9. A **dyadic product** or outer product is a type of multiplication that generates a second-order tensor called a **dyadic** from two vectors:

$$\overleftrightarrow{\mathbf{C}} = \mathbf{A}\mathbf{B}$$

Dyadics have nine independent components and in Cartesian coordinates can be decomposed into **dyads**:

$$\overleftrightarrow{\mathbf{C}} = A_x B_x \hat{\mathbf{x}}^2 + A_x B_y \hat{\mathbf{x}}\hat{\mathbf{y}} + A_x B_z \hat{\mathbf{x}}\hat{\mathbf{z}} + A_y B_x \hat{\mathbf{y}}\hat{\mathbf{x}} + A_y B_y \hat{\mathbf{y}}^2 + A_y B_z \hat{\mathbf{y}}\hat{\mathbf{z}} + A_z B_x \hat{\mathbf{z}}\hat{\mathbf{x}} + A_z B_y \hat{\mathbf{z}}\hat{\mathbf{y}} + A_z B_z \hat{\mathbf{z}}^2$$

Thus, dyadics can be represented in matrix form:

$$\overleftrightarrow{\mathbf{C}} = \begin{bmatrix} A_x B_x & A_x B_y & A_x B_z \\ A_y B_x & A_y B_y & A_y B_z \\ A_z B_x & A_z B_y & A_z B_z \end{bmatrix} = \begin{bmatrix} A_x & A_y & A_z \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \mathbf{A} \mathbf{B}^T$$

The transpose symbol is usually omitted in E&M.

A dyadic $\overleftrightarrow{\mathbf{C}}$ has two invariants, **spur** (called cusp in lecture) $|\overleftrightarrow{\mathbf{C}}|$ and **rotation vector** $\langle \overleftrightarrow{\mathbf{C}} \rangle$:

$$|\overleftrightarrow{\mathbf{C}}| = \sum_i^3 C_{ii}$$

$$\langle \overleftrightarrow{\mathbf{C}} \rangle = \hat{\mathbf{x}}(C_{yz} - C_{zy}) + \hat{\mathbf{y}}(C_{zx} - C_{xz}) + \hat{\mathbf{z}}(C_{xy} - C_{yx})$$

Dyadics commute in addition and scalar multiplication but not in inner products:

$$\begin{aligned} \overleftrightarrow{\mathbf{C}} + \overleftrightarrow{\mathbf{D}} &= \overleftrightarrow{\mathbf{D}} + \overleftrightarrow{\mathbf{C}} \\ a \overleftrightarrow{\mathbf{C}} &= \overleftrightarrow{\mathbf{C}} a \\ \mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} &\neq \overleftrightarrow{\mathbf{C}} \cdot \mathbf{A} \end{aligned}$$

Definition 1.10. Every dyadic $\overleftrightarrow{\mathbf{C}}$ has a **conjugate** $\overleftrightarrow{\mathbf{C}}^*$ such that:

$$\mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} = \overleftrightarrow{\mathbf{C}}^* \cdot \mathbf{A}$$

In three dimensional Cartesian coordinates, the conjugate of a dyadic is simply its transpose.

1.2 Differential Calculus

Definition 1.11. The **gradient** of a three dimensional scalar-valued function $f(x, y, z)$ is a vector field that points in the direction of maximum increase of f and has a magnitude equal to the rate of increase:

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{d}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$$

Definition 1.12. The **divergence** of a three dimensional vector-valued function $f(x, y, z) = f_x \hat{\mathbf{x}} + f_y \hat{\mathbf{y}} + f_z \hat{\mathbf{z}}$ is a scalar that represents how much the vector field diverges or converges, analogous to sources and sinks in fluid flow:

$$\nabla \cdot f = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

Definition 1.13. The **curl** of a three dimensional vector-valued function $f(x, y, z) = f_x \hat{\mathbf{x}} + f_y \hat{\mathbf{y}} + f_z \hat{\mathbf{z}}$ is a vector field with magnitude representing the infinitesimal circulation, analogous to rotation in a fluid, and direction perpendicular to that circulation:

$$\nabla \times f = \left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \right) \hat{\mathbf{z}}$$

Gradient, divergence, and curl can all be found by treating the vector operator ∇ **del** as a vector:

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

Gradient, divergence, and curl behave like ordinary derivatives under summation and multiplication by scalars. Since gradient, divergence, and curl can be multiplied by scalars and vectors in multiple

ways, there are six product rules:

$$\begin{aligned}
 \nabla(fg) &= f\nabla g + g\nabla f \\
 \nabla(\mathbf{A} \cdot \mathbf{B}) &= \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\
 \nabla \cdot (f\mathbf{A}) &= f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \\
 \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\
 \nabla \times (f\mathbf{A}) &= f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f) \\
 \nabla \times (\mathbf{A} \times \mathbf{B}) &= (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})
 \end{aligned}$$

Since you can combine the three derivatives in various ways, there are five species of second derivatives, but all but two are either degenerate or are composed of the others:

$$\begin{aligned}
 \nabla \cdot (\nabla f) \\
 \nabla(\nabla \cdot f)
 \end{aligned}$$

Definition 1.14. The **Laplacian** of three dimensional scalar-valued function f is:

$$\nabla^2 f = \nabla \cdot (\nabla f)$$

1.3 Integral Calculus

Definition 1.15. A **line integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal displacement along a curve:

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}$$

Definition 1.16. A **surface integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal area over a surface:

$$\int_S \mathbf{v} \cdot d\mathbf{a}$$

Definition 1.17. A **volume integral** of a scalar-valued function is the integral of the function over a volume:

$$\int_v T d\tau$$

Since there are three types of derivatives in three dimensions, there are three versions of the fundamental theorem of calculus.

Theorem 1.1. Gradient Theorem

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Proof. left as an exercise for the reader (will do tonight) □

Theorem 1.2. Divergence Theorem or Gauss's Theorem

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_{\partial V} \mathbf{v} \cdot d\mathbf{a}$$

Proof. left as an exercise for the reader (will finish tonight) □

Theorem 1.3. Stoke's Theorem

$$\int_S (\nabla \times \mathbf{v}) d\mathbf{a} = \oint_{\partial S} \mathbf{v} \cdot d\mathbf{l}$$

Proof. left as an exercise for the reader (will finish tonight) □

1.4 Curvilinear Coordinates

Definition 1.18. Spherical coordinates are a coordinate system that uses distance from the origin r , the angle down from the z-axis called the **polar angle** θ , and the angle around from the x-axis called the **azimuthal angle** ϕ .

The following are true for spherical coordinates:

- Conversion to Cartesian coordinates: $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$
- Unit vectors in terms of cartesian coordinates:

$$\hat{\mathbf{r}} = \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}}$$

$$\hat{\theta} = \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}}$$

$$\hat{\phi} = -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}$$

- Infinitesimal displacement: $d\hat{\mathbf{l}} = dr\hat{\mathbf{r}} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$
- Infinitesimal volume: $d\tau = r^2 \sin \theta dr d\theta d\phi$
- Gradient: $\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\phi}$
- Divergence: $\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$
- Curl: $\nabla \times \mathbf{v} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi} \right] \hat{\mathbf{r}} + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi) \right] \hat{\theta} + \frac{1}{r} \left[\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta} \right] \hat{\phi}$
- Laplacian: $\nabla^2 T = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial T}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2}$